NOTENOVA

LEARN SMART • REVISE BETTER • AIM HIGHER

CLASS 10 | SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

CHAPTER 1 STUDY PACK

INSIDE THIS PACK

CHAPTER NOTES	EXAM PRACTICE	WORKSHEET
Definitions, equations, reaction types, observations	Important short and long questions	MCQs, balancing, reaction identification + answer key

Easy English notes with Hinglish explanations • Original supplementary study material

1. Chemical Reaction

Definition: A chemical reaction is a process in which one or more substances change into new substances with different properties.

HINGLISH EXPLANATION
Jab ek ya zyada substances react karke naye substances banate hain, us process ko chemical reaction kehte hain.

$2Mg + O_2 \rightarrow 2MgO$

Magnesium reacts with oxygen to form magnesium oxide. Magnesium and oxygen are the reactants; magnesium oxide is the product.

Signs of a Chemical Reaction

A reaction may show one or more of these signs: (1) colour change, (2) temperature change, (3) evolution of gas, (4) formation of a precipitate, or (5) change in state.

IMPORTANT
A reaction does not have to show all these signs every time.

2. Chemical Equations

A chemical equation represents a chemical reaction using symbols and formulae.

Word equation

Magnesium + Oxygen → Magnesium oxide

Skeletal equation

$Mg + O_2 \rightarrow MgO$

This is unbalanced because the number of oxygen atoms differs on the two sides.

Balanced equation

$2Mg + O_2 \rightarrow 2MgO$

Term	Meaning
Reactants	Substances that take part in a reaction.
Products	New substances formed in a reaction.
Balanced equation	Each element has an equal number of atoms on both sides.
Precipitate	An insoluble solid formed when suitable solutions react.

LAW OF CONSERVATION OF MASS
Mass can neither be created nor destroyed in a chemical reaction. This is why chemical equations must be balanced.

3. How to Balance an Equation

Use the hit-and-trial method. Balance atoms by changing coefficients placed before formulae. Never change the subscripts inside a chemical formula.

Example

$$Fe + H_2O \rightarrow Fe_3O_4 + H_2$$

Step 1: Put 3 before Fe to balance iron.
Step 2: Put 4 before H_2O to balance oxygen.
Step 3: Put 4 before H_2 to balance hydrogen.

$$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$$

QUICK CHECK
Left: Fe = 3, O = 4, H = 8. Right: Fe = 3, O = 4, H = 8. The equation is balanced.

4. Symbols in Chemical Equations

Symbol	Meaning
(s)	Solid
(l)	Liquid
(g)	Gas
(aq)	Aqueous solution
Δ	Heat supplied
↑ / ↓	Common notation for gas evolved / precipitate formed

$$CaCO_3(s) \xrightarrow{heat} CaO(s) + CO_2(g)$$

5. Types of Chemical Reactions

A. Combination Reaction

Two or more substances combine to form a single product. Hinglish: Multiple reactants, one product.

$$CaO + H_2O \rightarrow Ca(OH)_2 + heat$$

$$2Mg + O_2 \rightarrow 2MgO$$

B. Decomposition Reaction

One compound breaks down into two or more simpler substances.

$$CaCO_3 \xrightarrow{heat} CaO + CO_2$$

$$2H_2O \xrightarrow{electricity} 2H_2 + O_2$$

$$2AgCl \xrightarrow{sunlight} 2Ag + Cl_2$$

Types: Thermal decomposition (heat), electrolytic decomposition (electricity), and photochemical decomposition (light). Silver chloride changes from white to grey in sunlight because silver forms.

C. Displacement Reaction

A more reactive element displaces a less reactive element from its compound.

$$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$$

Observation: The blue copper sulphate solution becomes greenish, and copper is deposited.

D. Double Displacement Reaction

Ions are exchanged between two compounds.

$$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$$

A white precipitate of barium sulphate forms.

E. Precipitation Reaction

An insoluble solid forms when suitable solutions react.

$$Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$$

A yellow precipitate of lead iodide forms. Precipitation reactions are commonly double displacement reactions.

6. Exothermic and Endothermic Reactions

Type	Meaning	Example
Exothermic	Releases energy, usually as heat.	$CaO + H_2O \rightarrow Ca(OH)_2$ + heat
Endothermic	Absorbs energy from surroundings.	$CaCO_3 \xrightarrow{\text{heat}} CaO + CO_2$

Respiration is considered exothermic because energy is released as glucose is broken down in cells.

7. Oxidation, Reduction and Redox

Oxidation: Addition of oxygen or removal of hydrogen.
Reduction: Removal of oxygen or addition of hydrogen.

$$CuO + H_2 \rightarrow Cu + H_2O$$

Substance	Change	Conclusion
CuO	Loses oxygen to form Cu	Reduced; acts as oxidising agent
H_2	Gains oxygen to form H_2O	Oxidised; acts as reducing agent

REDOX REACTION
Oxidation and reduction occur simultaneously in a redox reaction. An oxidising agent causes oxidation and is itself reduced; a reducing agent causes reduction and is itself oxidised.

8. Corrosion

Definition: Corrosion is the gradual damage of a metal due to reactions with substances in its environment.

Examples: rusting of iron, a green coating on copper, and a black coating on silver.

Rusting of iron

Iron rusts in the presence of both oxygen and moisture. Rust is hydrated iron(III) oxide, commonly represented as:

$$Fe_2O_3 \cdot xH_2O$$

Prevention

Painting, oiling or greasing, galvanisation, electroplating, and using suitable corrosion-resistant alloys.

HINGLISH TIP
Metal ki surface ko air aur moisture se bachane par corrosion ko roka ya kam kiya ja sakta hai.

9. Rancidity

Definition: Rancidity is the oxidation of fats and oils in food, causing an unpleasant smell and taste.

Prevention: (1) store food in airtight containers, (2) refrigerate when appropriate, (3) keep food away from excessive heat and light, (4) use suitable antioxidants, and (5) pack certain foods with nitrogen gas.

Nitrogen is used in packets of some chips to reduce oxidation of oils and fats.

10. Important Reaction Observations

Experiment	Observation	Equation
Burn magnesium ribbon	Bright white flame; white powder forms.	$2Mg + O_2 \rightarrow 2MgO$
Heat ferrous sulphate	Green crystals lose water; stronger heating forms a brown solid and gases.	$2FeSO_4 \xrightarrow{heat} Fe_2O_3 + SO_2 + SO_3$
Heat lead nitrate	Brown fumes of nitrogen dioxide form.	$2Pb(NO_3)_2 \xrightarrow{heat} 2PbO + 4NO_2 + O_2$
Silver chloride in sunlight	White solid turns grey.	$2AgCl \xrightarrow{sunlight} 2Ag + Cl_2$
Iron in $CuSO_4$	Blue solution becomes greenish; copper deposits.	$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
Na_2SO_4 + $BaCl_2$ solutions	White precipitate forms.	$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$

LAB SAFETY
Some reactions release toxic gases or involve corrosive substances. Perform experiments only in a properly equipped school laboratory under teacher supervision.

11. Important Exam Questions

Very Short Answer — 1 Mark

1. What is a chemical reaction?
2. What is a precipitate?
3. Define oxidation.
4. What is rancidity?
5. Why must a chemical equation be balanced?
6. Name the gas evolved when calcium carbonate is heated.
7. What is the colour of freshly formed silver chloride?
8. Name one method of preventing rusting.

Short Answer — 2–3 Marks

9. Write three observations that indicate a chemical reaction has occurred.
10. Differentiate between combination and decomposition reactions with examples.
11. What is a displacement reaction? Give one balanced equation.
12. Explain the difference between oxidation and reduction.
13. Why does copper sulphate solution change colour when iron is added?

14. What are exothermic and endothermic reactions? Give one example of each.

15. Why is respiration considered an exothermic process?

16. State two methods of preventing rancidity.

Long Answer — 3–5 Marks

17. Explain four major types of chemical reactions with balanced equations.

18. What is a redox reaction? Explain oxidation, reduction, oxidising agent and reducing agent using an equation.

19. What is corrosion? Explain rusting and describe three ways to prevent corrosion.

20. Explain why equations must be balanced and balance an example using the hit-and-trial method.

12. Worksheet — Practice Time!

INSTRUCTIONS
Attempt without looking at the notes. Write formulae correctly and balance all equations.

Section A — Multiple Choice Questions

1. Which is a combination reaction?
A. $CaCO_3 \rightarrow CaO + CO_2$ B. $2Mg + O_2 \rightarrow 2MgO$
C. $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ D. $2AgCl \rightarrow 2Ag + Cl_2$

2. Decomposition of AgCl in sunlight is: A. Combination B. Displacement C. Photochemical decomposition D. Neutralisation

3. Which is the reducing agent in $CuO + H_2 \rightarrow Cu + H_2O$? A. CuO B. H_2 C. Cu D. H_2O

4. Which gas is released when calcium carbonate is heated? A. O_2 B. H_2 C. CO_2 D. N_2

5. Rancidity is mainly associated with oxidation of: A. Water B. Fats and oils C. Salt D. Sand

Section B — Balance the Equations

6. $H_2 + O_2 \rightarrow H_2O$

7. $Al + O_2 \rightarrow Al_2O_3$

8. $Zn + HCl \rightarrow ZnCl_2 + H_2$

9. $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + NaCl$

10. $Fe + H_2O \rightarrow Fe_3O_4 + H_2$

Section C — Identify the Reaction Type

11. $CaO + H_2O \rightarrow Ca(OH)_2$

12. $2Pb(NO_3)_2 \xrightarrow{\text{heat}} 2PbO + 4NO_2 + O_2$

13. $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

14. $Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$

15. $CuO + H_2 \rightarrow Cu + H_2O$

Section D — Application-Based Questions

16. An iron nail is placed in blue copper sulphate solution. State two observations and explain them.

17. Oily food develops an unpleasant smell after being left open. Name the process and suggest two preventive measures.

18. Why does painting an iron gate help prevent rusting?

19. A white compound turns grey in sunlight. Identify a likely compound and write its balanced reaction.

20. In $CuO + H_2 \rightarrow Cu + H_2O$, identify the substance oxidised and the substance reduced.

13. Worksheet Answer Key

Section A

1. B 2. C 3. B 4. C 5. B

Section B

6. $2H_2 + O_2 \rightarrow 2H_2O$
7. $4Al + 3O_2 \rightarrow 2Al_2O_3$
8. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
9. $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$
10. $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

Section C

11. Combination reaction.
12. Thermal decomposition.
13. Displacement reaction.
14. Double displacement and precipitation reaction.
15. Redox reaction.

Section D — Key Points

16. Blue solution becomes greenish and reddish-brown copper deposits; iron displaces copper.
17. Rancidity; use airtight storage and refrigeration when appropriate.
18. Paint creates a protective barrier against air and moisture.
19. Silver chloride; $2AgCl \xrightarrow{\text{sunlight}} 2Ag + Cl_2$.
20. H_2 is oxidised; CuO is reduced.

14. Quick Revision

Topic	Remember
Combination	Two or more reactants form one product.
Decomposition	One compound forms two or more products.
Displacement	A more reactive element displaces a less reactive one.
Double displacement	Ions are exchanged between compounds.
Precipitation	An insoluble solid forms.
Oxidation	Addition of oxygen or removal of hydrogen.
Reduction	Removal of oxygen or addition of hydrogen.
Exothermic	Energy released.
Endothermic	Energy absorbed.
Corrosion	Gradual damage of metals.
Rancidity	Oxidation of fats and oils.

NOTENOVA STUDY TIP
Understand the reaction pattern first, then practise balancing. Do not memorise an unbalanced equation as if it were correct.

END OF CHAPTER 1 STUDY PACK

Prepared as original supplementary educational study material for NoteNova. This is not an official PSEB publication. Follow your prescribed textbook and current syllabus.